

How Does Climate Risk Affect Labor Mobility?

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② Literature Review

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Motivation

- Key question: how are labor movement choices made?
- Given the increasing prevalence of climate change, it is important to know to what extent climate risks impact labor mobility
- This has implications for how cities should manage their labor force and mitigate climate risks with other benefits to sustain their workforce.

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- Many papers looking at determinants of labor mobility (Ganesch, 2018; Greenwood, 1969; Gallaway et al., 1967)
 - Age
 - Agglomeration/peripheral regions
 - Qualifications of employees
 - Migrant stock (the number of people from state i living in state j)
 - Income differentials
 - Distance from origin to destination
- Song and Zhang (2024) found that temperature increases affect labor mobility in China. Mediating factors are environment regulations, individual happiness and health

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Data

- Aggregated labor data from Revelio
 - Test on Aug 2024 - Aug 2025 data (>25mil non-zero net labor mobility observations)
 - For each firm by month and qualification (seniority), sector/role, and metropolitan area: aggregate and external inflow and outflow, total employees, salary, amount of tasks that can be remote, average duration of employment
 - Processed by metropolitan area and month: Inflow, outflow and net inflow; Change in salary and skills
 - Layoff data by firm and month
- Climate data
 - National Risk Index: climate risk score determined by expected annual loss, social vulnerability, and community resilience (data not available regularly, four periods from 2020)
 - EM-DAT data on natural disasters: at the state/county level, 190 observations 2020-2021; 54 observations in 2024-2025

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Analysis

TWFE Regression model

$$Y_{it} = \beta' ClimateRisk_{it} + TFE_t + MetroFE_i + \gamma X_{it} + \zeta_{it}$$

Analysis is on the metropolitan area level (aggregated from firm)

Climate Risk:

- Breaking down the National Risk Index:
 - Social Vulnerability
 - Community Resilience
 - Natural Risk Hazard
- Occurance of natural disaster (EM-DAT)

Y_{it} : net inflow of workers, salary and skills interacted with inflow of workers

X_{it} : control variables e.g. housing prices, demographic data, crime, cost of living, unemployment rates etc. (political leanings?)

Heterogeneity: identify CATE of climate risk on labor mobility by

Other Thoughts

- Spatial effects: cluster hotspots and coldspots - literature has found geographically heterogeneous climate effects
- Analysis with lagged net inflows
- Robustness checks
 - Matching on controls
 - Causal forests for heterogeneity analysis

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Limitations

- Data scraping and prediction process unclear: salaries, inflows and outflows are predicted. How accurate is the prediction model?
- Range of aggregation period and lagged variables might make a big difference
- No individual or directional mobility/migration data

References

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